

EXP. NO: 2

EDA - Data Import and Export

AIM:

To import data from CSV, Excel, and web sources using Pandas, handle different data formats, and export a DataFrame to an Excel file.

PROCEDURE:

1. Import the Pandas library.
2. Read the CSV file using `pd.read_csv()`.
3. Read the Excel file using `pd.read_excel()`.
4. Display the first few records using `head()`.
5. Import tabular data from a webpage using `pd.read_html()`.
6. Select the required table from the webpage.
7. Store the data in a Pandas DataFrame.
8. Export the DataFrame to an Excel file using `to_excel()`.
9. Verify the exported data using `head()`.

PROGRAM:

```
import pandas as pd
df_csv = pd.read_csv("/content/test_Y3wMUE5_7gLdaTN.csv")
print("CSV Data:")
print(df_csv.head())
df_excel = pd.read_excel(
    "/content/Historicalinvesttemp.xlsx",
    sheet_name="Sheet1")
print("\nExcel Data:")
print(df_excel.head())
url = "https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal)"
tables = pd.read_html(url)
df_web = tables[0]
print("\nWeb Data:")
print(df_web.head())
df = pd.read_excel("/content/Historicalinvesttemp.xlsx")
df.to_excel("/content/weeeeee.xlsx", index=False)
print("\nData exported successfully!")
print(df.head())
```

OUTPUT:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6
0	NaN	Excel Sample Data	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	Students Marksheet Data	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	Total Marks	300

	Student Name	Exam Name	Date	Points
0	James Walker	Biology	05/06/2016	31
1	Velma Clemons	Biology	05/06/2016	43
2	Kibo Underwood	Biology	05/06/2016	81
3	Louis Mcgee	Biology	05/06/2016	11
4	Phyllis Paul	Biology	05/06/2016	18

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

RESULT:

Thus, the given program was successfully executed to import data from CSV, Excel, and web sources and export a DataFrame to an Excel file.